IOQM · GRADE 9 MATHEMATICS

Modular Arithmetic, Residues and Power Cycles

Diagnostic and coverage-hardening companion

TARGET MODULUS → REDUCE → CYCLE / THEOREM → LEGALITY → CHECK

Teacher-only companion - answer diagnostics and theorem-hypothesis checks

NT-02 - Teacher Diagnostic Key

Teacher/authoring material; not in the student PDF.

Recognition lab

1. $5 \mid (38-3) = 35$.
2. Not equal; congruent mod 12 because 12 divides 12.
3. $12345 \equiv 4 (mod 7)$.
4. 2,4,1,2.
5. Mod 10.
6. Mod 100.
7. Inverse 3 mod7 is 5: $x \equiv 5 \cdot 6 \equiv 2$.
8. gcd(2,6)=2, so 2 has no inverse mod6.
9. $x = 2 + 3k$, then test mod4.
10. Incompatible: residues disagree mod gcd(4,6)=2.
11. Avoid $m \mid (A - B)$.
12. Base mod7 and exponent mod6 (for nonzero base states).
13. m divides a-b.
14. 5 inverse mod12 is 5.
15. No: gcd(6,15)=3.
16. $a^2 \equiv b^2 (mod 9)$.

Practice answers

1. 6,1,1,1.
2. difference 30 divisible by10.
3. 3.
4. 0,1,2,3,4.
5. cycle length3; $40 mod 3 = 1$, answer2.
6. 7 powers mod10 cycle 7,9,3,1; $2026 mod 4 = 2$, answer9.
7. 25.
8. inverse3 mod7=5; $x \equiv 25 congruent 4$.
9. $x \equiv 8 (mod 15)$.
10. 3 has order6; $100 mod 6 = 4$; $3^4 = 81 congruent 4$.
11. 2^{2025} last digit2; 3^{2025} last digit3; sum last digit5.
12. inverse5 mod12=5; x congruent40 congruent4.
13. divide condition correctly: 4(x-2) divisible12 -> x congruent2 (mod3); classes 2,5,8,11 mod12.
14. gcd(6,9)=3; difference3 divisible3, compatible.
15. numbers 3 mod4: 3,7,11,15,19,...; first 5 mod7 is19.
16. $3^{20} mod 100 = 1$, so last two digits01.
17. induction from 25*5=125 congruent25 mod100.
18. 3.
19. 6.
20. polynomial operations preserve congruence.
21. $2 \cdot 1 congruent 2 \cdot 5 (mod 8)$ but 1 not congruent 5 mod8.

22. $6x = 9(mod 15)$ has no solution because gcd(6,15)=3 divides9; divide by3 gives `2x congruent3 (mod5)`, x congruent4 mod5 -> classes `4,9,14 mod15`.
23. $2^{1000} mod 7 = 2$, $3^{1000} mod 7 = 4$; total6.
24. $11^2 = 121 congruent 21$, powers mod100 cycle? Direct binomial $(1+10)^n \equiv 1 + 10n (mod 100)$; n=2026 gives61.
25. gcd(6,9)=3; residue difference3 divisible3, compatible. $x = 4 + 6k$; k congruent? 4+6k congruent7 mod9 ->6k congruent3 ->2k congruent1 mod3 ->k congruent2 mod3; least x=16.
26. `7(x-2)` divisible21 -> `x congruent2 (mod3)`; residues `2,5,8,11,14,17,20`.
27. 31.
28. preserve zero set ->7|T; preserve all nonzero base powers ->3^T=1 mod7, order6 ->6|T.
29. gcd(10,30)=10; cancellation only reduces modulus to3; conclusion is `x congruent2 (mod3)`.
30. gcd(4,6)=2 but residue difference3 not divisible2: no solution.

H0 answers

1. $12 \mid (83 - 11)$.
2. Work mod10; cycle of 7.
3. gcd(4,9)=1; inverse exists.
4. gcd(6,9)=3; difference3 is divisible3, so compatible.
5. $2026 mod 3 = 1$; answer2.
6. 25.
7. inverse5 mod12=5; x congruent35 congruent11.
8. gcd(6,18)=6 divides12; divide by6: x congruent2 mod3 -> classes `2,5,8,11,14,17`.
9. 9.
10. $1234 mod 6 = 4$; answer $3^4 mod 7 = 4$.
11. gcd(4,6)=2 but residue difference1 not divisible2: no solution.
12. powers of7 mod10 cycle length4; exponent $7^7 mod 4 = 3$; answer $7^3 mod 10 = 3$.
13. `14|42` is equivalent to `42 congruent0 (mod14)`; divisibility is the zero-residue special case of congruence.
14. Mod7: inverse3 exists ->x congruent2. Mod9: gcd3,9=3; reduce to `x congruent2 (mod3)` -> classes 2,5,8 mod9.
15. 31; below 14 pigeonhole, 14..30 each has a collision, mod31 fourth-power residues are all distinct.
16. $2^n mod 7$ cycles `2,4,1`; $100 mod 3 = 1$; residue2.

Diagnostic tags

- `NT02-R1` equality/congruence confusion.
- `NT02-R2` divisibility/congruence bridge missing.
- `NT02-R3` wrong target modulus.
- `NT02-R4` brute-force power instead of cycle.
- `NT02-R5` illegal cancellation/inverse gap.
- `NT02-R6` simultaneous-congruence compatibility gap.
- `NT02-R7` base-period/exponent-period confusion.

NT-02 - Teacher Coverage Hardening Addendum

Status: `STATIC_DIAGNOSTIC_ADDENDUM_V1` Issue: #132

This addendum synchronizes the learner-facing bounded Euler bridge in `03_First_Step_Reference.md`. It does not alter historical-anchor answers or existing authored-item keys.

Euler's theorem

If $gcd(a, n) = 1$, then

$a^{\varphi(n)} \equiv 1 (mod n)$,

where $\varphi(n)$ counts the integers in `1,...,n` that are coprime to n.

Independent proof-idea check

Let $r_1, ..., r_phi$ be a complete list of invertible residue classes modulo n. Multiplication by a permutes these classes because a is invertible. Therefore

$(ar_1)...(ar_phi) \equiv r_1...r_phi(mod n)$.

The product $r_1...r_phi$ is invertible, so cancellation is legal, yielding

$a^{\varphi(n)} \equiv 1 (mod n)$.

Decision boundary

A large exponent does not automatically call for Euler. First reduce the base and inspect a short cycle/order. Use Euler only when $gcd(a, n) = 1$ is explicit or cheaply checked and the theorem reduces work.

Fermat's little theorem - design companion

For prime p with p not dividing a,

$a^{p-1} \equiv 1 (mod p)$.

This is included as a curriculum-design prime-modulus companion/corollary. The supplied preparation routine explicitly names Euler's theorem, not Fermat's little theorem.

Diagnostic codes

- `NT02-EULER-1`: coprimality not checked before Euler.
- `NT02-EULER-2`: exponent reduced by $\varphi(n)$ in a non-coprime case.
- `NT02-EULER-3`: totient machinery overused where a two- or three-step residue cycle is cheaper.
- `NT02-FERMAT-1`: modulus not checked prime.
- `NT02-FERMAT-2`: base divisible by prime modulus but Fermat invoked anyway.
- `NT02-POWER-ROUTE`: theorem/cycle choice not justified.

Evidence truth

This addendum is a static mathematics/diagnostic synchronization. It does not constitute classroom timing/readability, retention, psychometric, qualification/pass-mark or publication calibration; those remain `NOT_RUN`.